TELECOMMUNICATION LAB COMPLETE EXAM PREPARATION

Experiments 1, 6, 7 and 12 | MATLAB + Python scripts | Expected plots | Viva answers

Use this guide in a 3-hour window. Learn the aim, transmitter/receiver sequence, core code, plotted outputs and 8-12 viva answers for each experiment. The `.m` and `.py` programs are included separately in the ZIP.

Three-hour study plan

Time	Task	What to be able to say
0-10 min	Open files + run Exp 1	Know where code and expected PNGs live.
10-65 min	Exp 1 - PCM	Sampling, quantization, bit mapping, LPF, six plots.
65-100 min	Exp 6 - DSSS	PN chips, spreading and synchronized despreading.
100-135 min	Exp 7 - FHSS	Seven hopping frequencies and the dehopping idea.
135-165 min	Exp 12 - codes	XOR connections and 6 output pairs.
165-180 min	Viva rehearsal	Explain every experiment without looking.

What was wrong with the old ZIP?

The original Python files run. Most programs are simply too long for a practical exam. Exp 1 uses a moving-average reconstruction rather than the manual's Butterworth LPF. Exp 6 uses a longer data message and adds optional BER results; Exp 7 includes alternative lengthy versions; Exp 12 prints much more than necessary. The original ZIP contains no MATLAB .m scripts.

What is inside the new ZIP?

MATLAB/: Four readable .m scripts. **Python/:** Four short .py scripts. **Expected_Outputs/:** Four verified PNG waveform images. **Quick_Revision_One_Page.pdf:** Morning recap. **REVIEW_OF_YOUR_OLD_ZIP.md:** Specific corrections and manual differences.

Software note: New Python scripts were executed successfully. MATLAB itself was not installed in the testing environment, so the MATLAB scripts are supplied as carefully reviewed equivalents, not claimed as MATLAB-runtime tested. The PCM scripts use Butterworth where optional signal-processing functions are available; otherwise a first-order low-pass fallback.

EXPERIMENT 1 | PCM Modulation & Demodulation

Lab manual PDF pages 2-4 | Aim: Generate Pulse Code Modulation and Demodulation.

One-line explanation: PCM converts analog samples into discrete quantized values, assigns each a binary word, then decodes and low-pass filters them to reconstruct an approximation of the input.

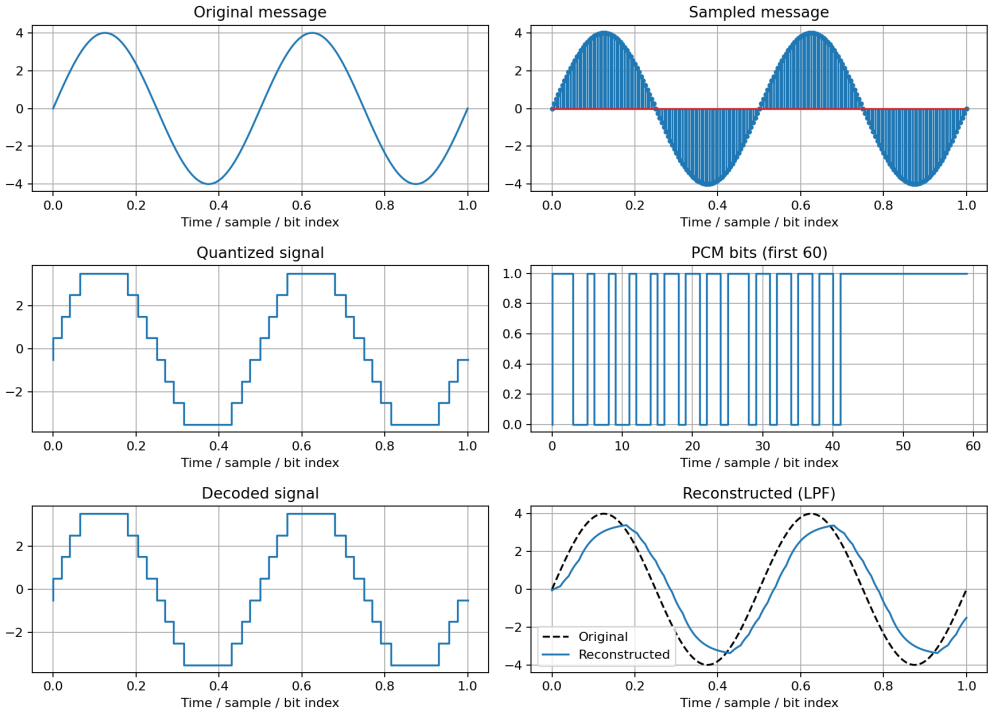
Working sequence

Transmitter: message -> anti-alias low-pass filter -> sampler -> quantizer -> 3-bit encoder -> PCM bits. **Receiver:** regenerated PCM bits -> decoder -> reconstruction low-pass filter -> analog approximation.

Parameter	Value	Exam significance
Amplitude A	4	Message x(t) = 4 sin(2 pi x 2 x t).
Message frequency fm	2 Hz	Highest message frequency in this demo.
Sampling fs	200 Hz	fs = 100 fm, safely above Nyquist minimum.
Quantizer	8 levels	-3.5, -2.5, ..., 3.5; step Delta = 1.
Bits per sample	3	log2(8) = 3; 000 through 111.
PCM bit rate	600 bits/s	fs times 3 bits/sample.

Worked example: x = 2.2 is assigned the nearest allowed level 2.5, corresponding to index 6 (zero-based), so the PCM word is **110**. If input lies in [-4,4], quantization error magnitude is at most 0.5.

Expected waveforms



Output adds a binary PCM-bit subplot to the five waveforms in the manual. The reconstructed sine is approximate because quantization has already lost amplitude precision.

Viva: quick answers

1. What is PCM? Pulse Code Modulation: sampling, quantization and binary encoding of an analog signal.

2. What does sampling do? It makes time discrete. Here fs = 200 Hz for a 2 Hz message.

3. State the sampling theorem. Sampling frequency must be at least twice the highest message frequency: fs >= 2fmax.

4. Why is quantization needed? To represent a continuous sample amplitude using one of a finite number of levels.

5. What is quantization error? The difference between original sample and assigned level; maximum 0.5 here (within the allowed input range).

6. Why 3 bits per sample? There are L = 8 levels, and log2(8) = 3 bits.

7. What is the demo PCM bit rate? Rb = fs x bits/sample = 200 x 3 = 600 bits per second.

8. What is the function of the LPF? At the receiver, it smooths the decoded staircase and attenuates high-frequency components.

9. Why is output not identical to input? Quantization introduces irreversible approximation, and filtering can add transient/delay.

10. Difference between sampled and quantized? Sampled: continuous amplitude at discrete times. Quantized: discrete time and discrete amplitudes.

11. Why use an anti-aliasing filter? Before sampling, it limits signal bandwidth so high frequencies do not fold into the sampled band.

12. What do stem and subplot do? stem shows individual samples; subplot places several waveform plots in one figure.

What to do in the lab

Run Exp1_PCM.m or Exp1_PCM.py. Point to the quantized staircase and the bit plot. Change fm from 2 to 3 only if asked; explain that sampling must still satisfy fs >= 2 fmax. In the MATLAB script the manual's Butterworth filter is used if the Signal Processing Toolbox is available.

EXPERIMENT 6 | Direct Sequence Spread Spectrum

Lab manual PDF pages 18-20 | Aim: Implement DSSS.

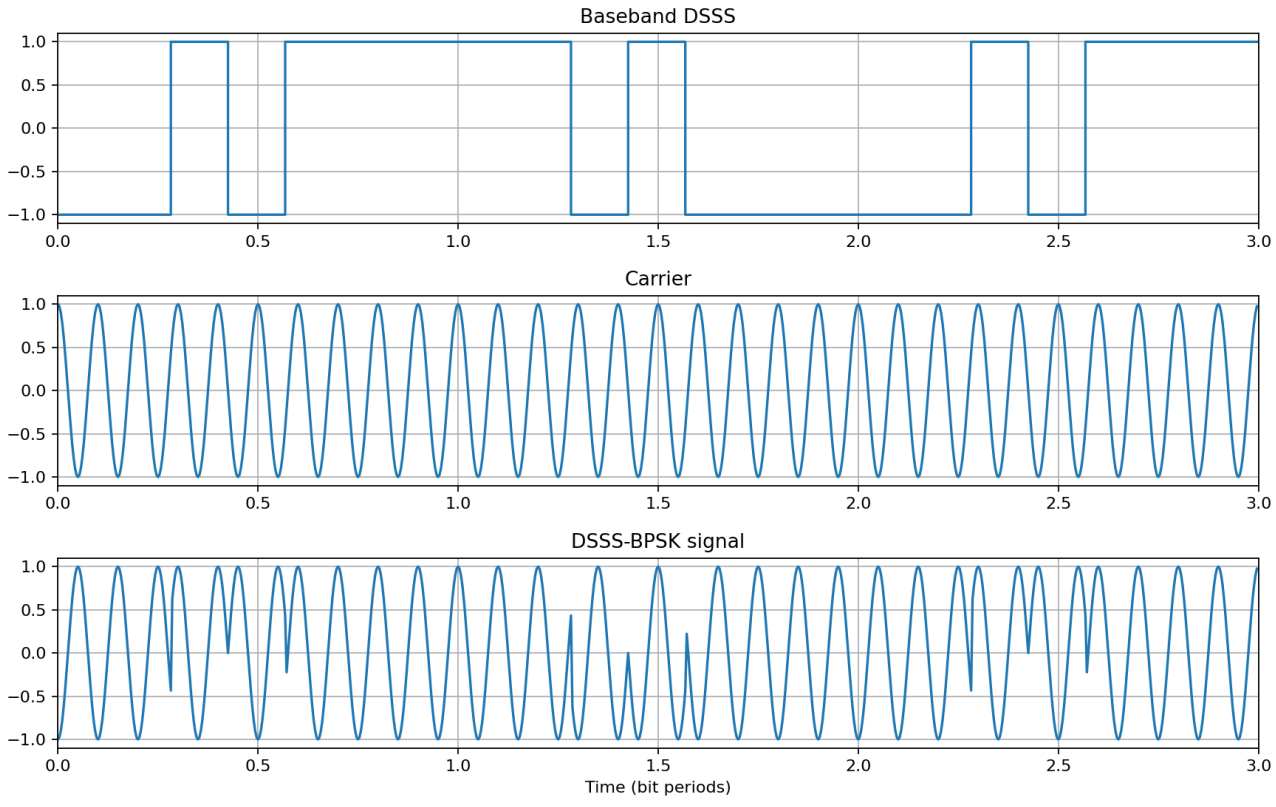
One-line explanation: Each input data bit is multiplied by a faster PN chip sequence. The resulting spread signal is BPSK-modulated; the synchronized receiver uses the same PN code to recover the original bits.

Transmitter and receiver

TX: data (0/1) -> bipolar NRZ (-1/+1) -> multiply by PN chips -> BPSK modulation -> channel. **RX:** synchronized carrier detection -> multiply by same PN waveform -> sum/correlate over one bit -> decide 1 for positive, 0 for negative.

Parameter	Manual example / new code
Input bits	1 0 1
PN sequence	0 0 1 0 1 1 1
Polar mapping	0 -> -1; 1 -> +1
Chips per bit	7; spread length = 3 x 7 = 21 chips
Processing gain	G = 7, or 10 log10 7 = 8.45 dB
Recovered bits	1 0 1 (ideal/noiseless, synchronized receiver)

Expected waveforms



Three plots match the manual emphasis: (1) baseband DSSS, (2) sinusoidal carrier, (3) transmitted BPSK-DSSS waveform. Recovery check is printed on the console.

Viva: quick answers

1. Expand DSSS. Direct Sequence Spread Spectrum.

2. What is the purpose of the PN code? A pseudo-noise chip sequence spreads each data bit and allows synchronized despreading.

3. Define a chip. One element of the spreading code; it is shorter in duration than an information bit.

4. What are the input bits and PN chips? Data = 101; PN = 0010111 (manual example).

5. What is the spreading factor? 7 chips per bit; chip rate is 7 times data bit rate.

6. Why BPSK? The bipolar spread baseband signal modulates the phase of a carrier using +1 or -1.

7. How does the receiver recover data? It uses the same synchronized PN sequence and carrier, then correlates over a bit interval.

8. What if the receiver PN is out of sync? Correlation is poor and the bit decisions may be incorrect.

9. What is processing gain here? 10 log10(7) = approximately 8.45 dB.

10. Advantage of DSSS? Interference resistance and code-based separation when properly synchronized.

Common confusion: Changing amplitude sign at PN-chip intervals represents a 180-degree BPSK phase reversal. The PN code does not need to be random every run; both ends must agree on it.

EXPERIMENT 7 | Frequency Hopping Spread Spectrum

Lab manual PDF pages 21-24 | Aim: Implement FHSS using MATLAB.

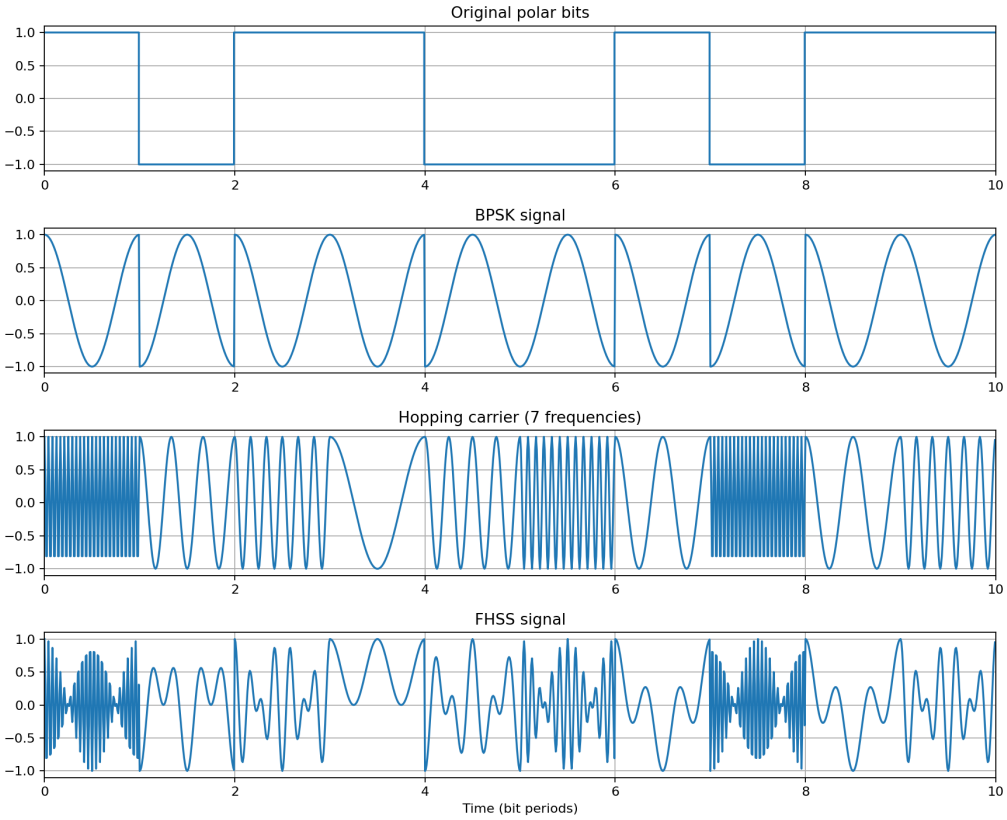
One-line explanation: FHSS changes the carrier frequency using a prearranged hopping sequence. With the matching local sequence, the receiver dehops and estimates the original data.

Transmitter and receiver

Manual diagram: binary data -> M-ary FSK modulator -> mixer with synthesizer (PN-controlled) -> FH/MFSK signal. **Manual program:** it actually constructs a BPSK waveform and multiplies it by a 7-frequency hopping carrier. The new code follows the program, not an invented replacement architecture.

Parameter	Exam-ready example
Input bits	1 0 1 1 0 0 1 0 1 1
Available frequencies	1, 2, 3, 4, 6, 12, 24 cycles/bit
Chosen hops	24, 3, 6, 1, 4, 12, 2, 24, 2, 6
Hopping selection	Fixed illustrative pattern, so the plots repeat
Received bits	1 0 1 1 0 0 1 0 1 1, assuming ideal synchronization

Expected waveforms



Four plots: original polar bits, BPSK signal, hopping carrier and FHSS output. Short dense oscillations are high-frequency hops. The multiplication in the manual creates sum/difference frequency components.

Viva: quick answers

1. Expand FHSS. Frequency Hopping Spread Spectrum.

2. Basic principle? The transmitter changes carrier frequency among selected channels under control of a hopping sequence.

3. What controls the hop? The PN/hopping code controls a frequency synthesizer.

4. Why is it spread spectrum? Signal energy is distributed over a wider overall band by successive carrier-frequency changes.

5. How is FHSS different from DSSS? DSSS spreads using high-rate PN chips; FHSS spreads by changing carrier frequency.

6. What does the receiver require? The same frequency set and a synchronized hopping sequence for dehopping.

7. How many hop frequencies does our experiment show? Seven available hopping frequencies.

8. What is the advantage against narrowband interference? Interference on one frequency may only affect hops that land on that frequency.

9. What are slow and fast hopping? Slow: one or more symbols per hop; fast: multiple hops within a symbol.

10. Why does the mixed waveform show multiple components? The manual multiplies a BPSK signal by a hopping carrier, creating sum and difference frequencies.

Manual compatibility warning: The older MATLAB code uses randint and Gray conversion; the prepared code intentionally avoids both. An ideal PN generator is not explicitly implemented; the fixed selection is an illustrative hop sequence.

EXPERIMENT 12 | Convolutional Codes

Lab manual PDF pages 35-36 | Aim: Implement a convolutional encoder in MATLAB.

One-line explanation: A convolutional encoder takes the current bit and memory of earlier bits, combines them using XOR gates, and generates two output bits per input bit.

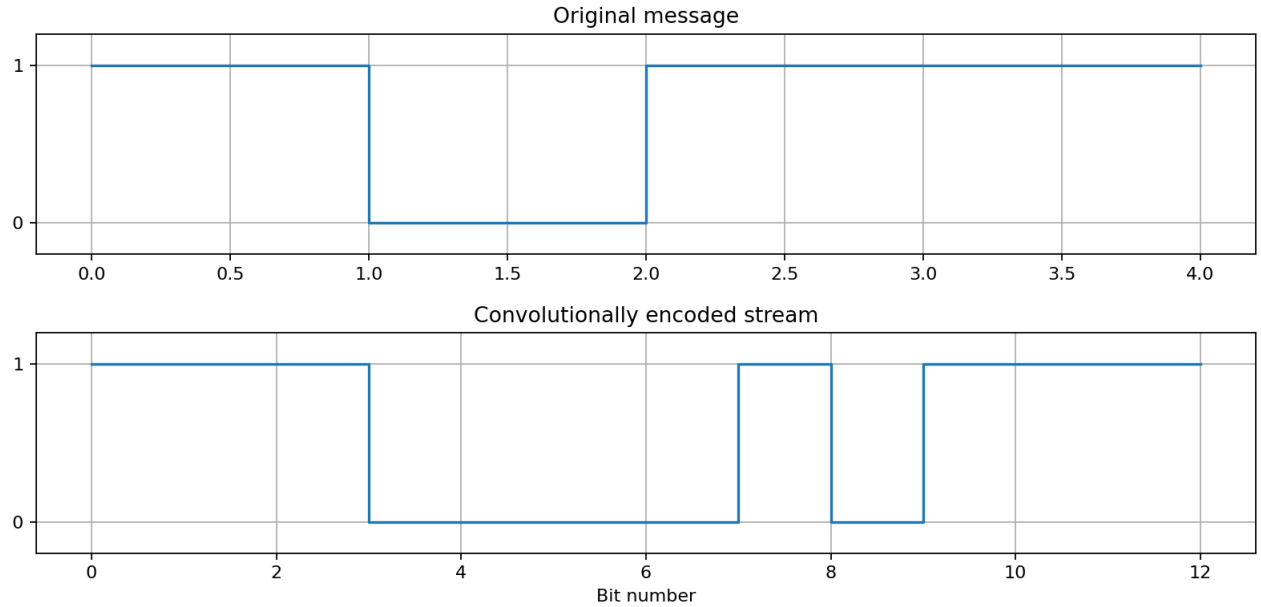
Encoder rule, taken from the manual program

Register state before a new bit: d1 and d2. For input bit u, first output c1 = u XOR d1 XOR d2, second c2 = u XOR d2. Shift by updating d2 <- d1 and d1 <- u. These are generator connections **111 and 101**.

Step	Input	Old state (d1d2)	Output (c1c2)
1	1	00	11
2	0	10	10
3	1	01	00
4	1	10	01
5	0 (tail)	11	01
6	0 (tail)	01	11

Must memorize: Input 1011 + two termination zeros -> 101100. Output pairs = **11 10 00 01 01 11**. Serialized output = **111000010111**. Nominal rate = 1/2, two memory registers, constraint length K = 3.

Expected visualization



The manual displays only the code-vector matrix. The added 2-panel waveform is for understanding the input bits and transmitted serial encoded bits.

Viva: quick answers

1. What is a convolutional code? A forward error-correcting code whose outputs depend on the current bit and earlier stored bits.

2. What is (2,1,2)? Two output bits per input bit, one input stream, two memory elements (manual notation).

3. What is the code rate? Nominal rate k/n = 1/2.

4. What is constraint length? K = memory + 1 = 3.

5. What is the purpose of shift registers? To retain previous input bits and create memory in the encoder.

6. Which operation combines bits? XOR (modulo-2 addition).

7. What are the generator connections? g1 = 111, g2 = 101, inferred from the manual XOR equations.

8. Why add two zeros at the end? They flush the two delay elements back to zero (termination).

9. What is the output for 1011? 11 10 00 01 01 11; serial bitstream 111000010111.

10. Does this lab implement error correction decoding? No. The supplied Exp 12 program is an encoder only; Viterbi decoding is a related topic.

11. What is a trellis? A time-indexed graph of encoder states and allowed transitions.

12. Why are there 12 coded bits for 4 data bits? Four input bits plus two tail zeros make six encoder steps; two output bits per step = 12 bits.

Result statement

A (2,1,2) convolutional encoder was implemented. For message 1011 with two zero tail bits, it generated output pairs 11 10 00 01 01 11, agreeing with the XOR generator connections.

FINAL 15-MINUTE SELF-TEST

Say each answer aloud without looking, then inspect the answer printed below. If you hesitate, review only that experiment, not the full notebook.

PCM: What are the three operations? Sampling, quantization, binary encoding.

PCM: How many bits for eight levels? Three.

PCM: Why 600 bits/s in this simulation? 200 samples/s times 3 bits/sample.

PCM: Why do we need a reconstruction LPF? To smooth a decoded staircase by reducing high-frequency components.

DSSS: What does PN stand for? Pseudo-noise or pseudorandom noise.

DSSS: What are the seven chips used here? 0010111.

DSSS: Why synchronized PN at RX? It permits successful correlation/despreading.

DSSS: What is processing gain? About 8.45 dB for spreading factor 7.

FHSS: What is hopping? Switching carrier frequency according to a code-controlled schedule.

FHSS: What must RX know? Exact hop frequencies, sequence and timing.

FHSS: Why narrowband interference resistance? A jammer usually affects only a subset of hops.

FHSS: What is the final plot? The frequency-hopped spread-spectrum waveform.

Convolutional: How many memory cells? Two.

Convolutional: What XOR equations? c1 = u XOR d1 XOR d2; c2 = u XOR d2.

Convolutional: What is 1011 encoded as? Output 11 10 00 01 01 11 after adding two zeros.

Convolutional: What is constraint length? Three; memory length 2 plus current input bit.

Exam practical: what to demonstrate

1. State aim in one sentence. **2.** Name blocks in the transmitter and receiver. **3.** Execute the correct script. **4.** Show one important numerical result. **5.** Explain each plot in order. **6.** Answer two or three viva questions.

Before sleep: Open the one-page sheet and look at the images once. Prioritize comprehension over memorizing 100+ lines of the earlier notebooks.